

Tridib Banik

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Education

Bachelor of Engineering, Software Engineering Co-op

McMaster University, Hamilton, ON

Sept. 2023 – Apr. 2028

CGPA: 11.32 / 12.0

- **Relevant Courses:** Data Structures & Algorithms (Python), Operating Systems (C), Databases (SQL), OOP (Java), Software Requirements & Security, Large System Design (UML), Dynamic Systems and Control (MATLAB/Simulink), Signals and Systems.

Skills

Languages: Python, Java, C, SQL, HTML, CSS, Bash, MATLAB.

Technologies: Flask, React, PostgreSQL, Docker, OpenAI Gymnasium, Stable-Baselines3, PyTorch, Simulink.

Tools: Git, GitHub, GitLab CI/CD, Pytest, JUnit, Maven, Linux, UML.

Work Experience

Research Assistant, CMHT Lab, McMaster Automotive Resource Centre

June 2025 – Aug. 2025

- Co-authored research paper (under review) profiling memory usage of a safety-critical electric car battery management system (BMS) on an automotive-grade microprocessor; benchmarked 7 subsystems, finding BMS logic uses only 0.99% flash and 9.23% RAM, with State of Charge alone consuming 54.03% and 56.72% of those totals.
- Built 4 model-based control logic modules (Contactor Control, Cell Balancing, AC/DC Fast Charging, Fault Diagnosis) in Simulink, verified against formal software requirements specifications.

BWC Software Team Member & Lab Assistant, CMHT Lab

Sept. 2024 – Present

- Automated Simulink model verification by configuring a 5-stage GitLab CI/CD pipeline; registered a GitLab Runner, managed CI variables, and documented setup for the team.
- Achieved 95%+ unit test pass rate across 100% of assigned subsystem requirements; contributed to release notes, system diagrams, and MATLAB signal figures; attended 24+ team lead meetings over 9 months covering budget allocation, deliverable planning, and risk mitigation.

Teaching Assistant, ENGINEER 1P13: Integrated Cornerstone Design Projects course

Jan. 2026 – Apr. 2026

- Mentored 23 students across 5 teams through a semester-long material selection project spanning 6 milestones and a final research presentation, delivering live verbal feedback to every team and individual on objective trees, metrics, regulatory compliance, and decision matrices for wastewater filtration systems.

Projects

TrafficLightRL | Python, OpenAI Gymnasium, Stable-Baselines3, PyTorch

Jan. 2025 – Mar. 2025

- Built a modular RL pipeline with separated environment, training, and evaluation components; trained a PPO agent across 3 traffic densities, achieving up to 29.55% lower vehicle wait time and 35.56% lower emissions vs. a fixed-cycle baseline; presented at the Canadian Undergraduate Conference on AI (320+ attendees).

Smart City Environmental Monitoring & Alert System | Software Architecture & Design

Jan. 2026 – Apr. 2026

- Designed a heterogeneous architecture (PAC overall; Pipe-and-Filter, Blackboard, and Repository across 3 subsystems) for a smart city platform; produced Analysis Class Diagrams, CRC cards, and detailed class diagrams across 4 structured deliverables.

CNC Data Pipeline | Python, Flask, React, PostgreSQL, Docker, Scikit-learn

Apr. 2026

- Built an Industry 4.0 analytics platform ingesting simulated Fanuc, Haas, and Mitsubishi CNC telemetry into PostgreSQL via Flask REST APIs; tracked OEE/KPI metrics including spindle load, tool wear, cycle time, and alarms.
- Developed a React dashboard (Recharts) for real-time and historical visualization with JWT authentication and role-based access control; implemented anomaly detection and predictive maintenance using Scikit-learn IsolationForest.